

FACTORS INFLUENCING PERFORMANCE, MOTIVATION AND STRESS LEVELS OF SOFTWARE PROFESSIONALS IN REMOTE WORK: A LEADERSHIP ROLE ANALYSIS

Authors: Camilo Daza, Kelly Garcés

Affiliations: Universidad de los Andes, School of Engineering,

Department of Systems and Computing Engineering, Bogotá, Colombia



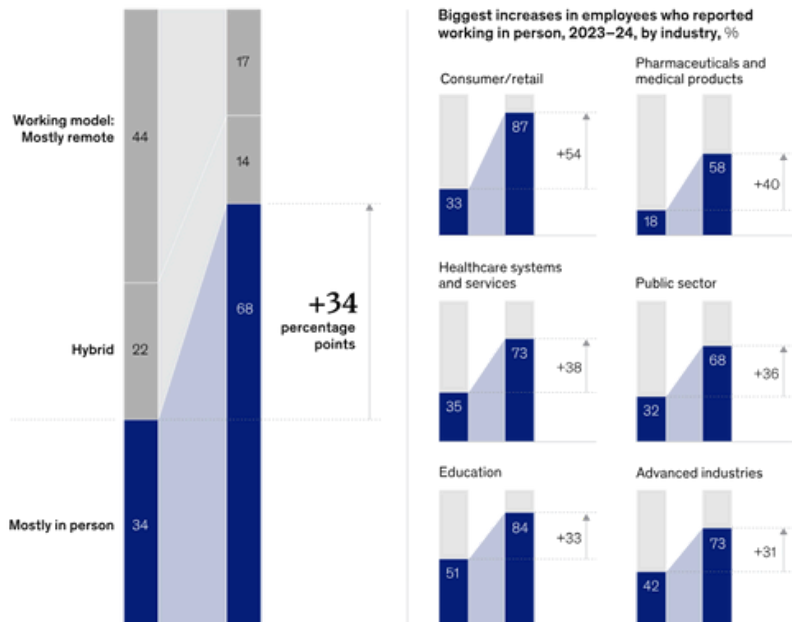
Introduction

The transition to remote work has fundamentally transformed workplace dynamics in software development, yet quantitative research examining leadership role differences remains limited in Iberoamerican countries. Organizations are struggling to understand the true impact of remote work, as evidenced by the significant shift back to in-person work arrangements across industries (see Figure 1). This uncertainty about remote work effectiveness has created substantial organizational challenges.

Figure 1: Back In-Person Work

More employees reported working in person in 2024 than in 2023.

Change in working model, 2023-24, %



Source: McKinsey & Company

Additionally, many organizations experience tensions between individual contributors preferring remote work and managers preferring return-to-office mandates.

Research Question: How do perceptions of performance, stress, and motivation differ between managerial and individual contributor roles in remote work environments across Iberoamerican countries?

Methodology

Participants/Sample: 232 software professionals

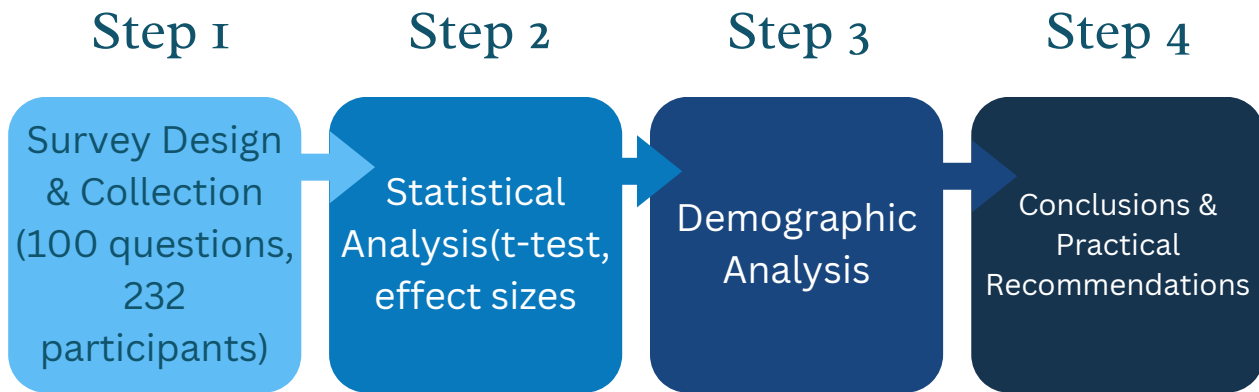
- 57 managers (24.8%) and 175 individual contributors (75.2%)
- Countries: Argentina, Colombia, Spain, Mexico
- Gender: 184 males (79.3%), 47 females (20.2%)

Instrument: 100-question survey measuring performance, motivation, and stress perceptions across remote vs. in-person work modalities

Procedure: Cross-sectional survey design comparing high remote work (83.6%) vs. low remote work (16.4%) participants

Data Analysis: T-tests, effect size calculations (Cohen's d), ANOVA and demographic interaction analysis

Figure 2. Research Methodology Steps



Results

Key Finding: Managers experience 3-4 times larger benefits from remote work compared to individual contributors.

Major Benefits for Managers in Remote Work:

- Team Milestones: +1.34 points ($p=0.0005$)
- Motivation: +1.30 points ($p<0.0001$)
- Problem Solving: +1.30 points ($p=0.0013$)
- Team Productivity: +1.23 points ($p=0.0135$)
- Less Coordination Issues: -1.04 points ($p=0.0053$)
- Less Communication Issues: -0.94 points ($p=0.0003$)

Modest Benefits for Individual Contributors:

- Comfort: +0.59 points ($p=0.0030$)
- Autonomy: +0.39 points ($p=0.0115$)
- Performance: +0.37 points ($p=0.0256$)
- Creativity: +0.34 points ($p=0.0366$)

Gender-Leadership Interactions:

- Male managers show strongest positive effects across all metrics
- Female individual contributors report decreased team productivity (-0.22 points)
- Remote work particularly benefits males in leadership positions

Conclusion

Remote work affects leadership roles dramatically differently in Iberoamerican software development contexts. Managers derive substantial benefits in team coordination, motivation, and productivity while experiencing reduced workplace problems. Individual contributors show more modest gains focused on personal well-being.

Key Implications:

- Organizations should implement differentiated remote work policies tailored to specific leadership roles
- Male-female disparities in remote work benefits require targeted interventions
- Role-based remote work strategies are essential for fair outcomes

Figure 3: Remote work impact by Role across studied metrics

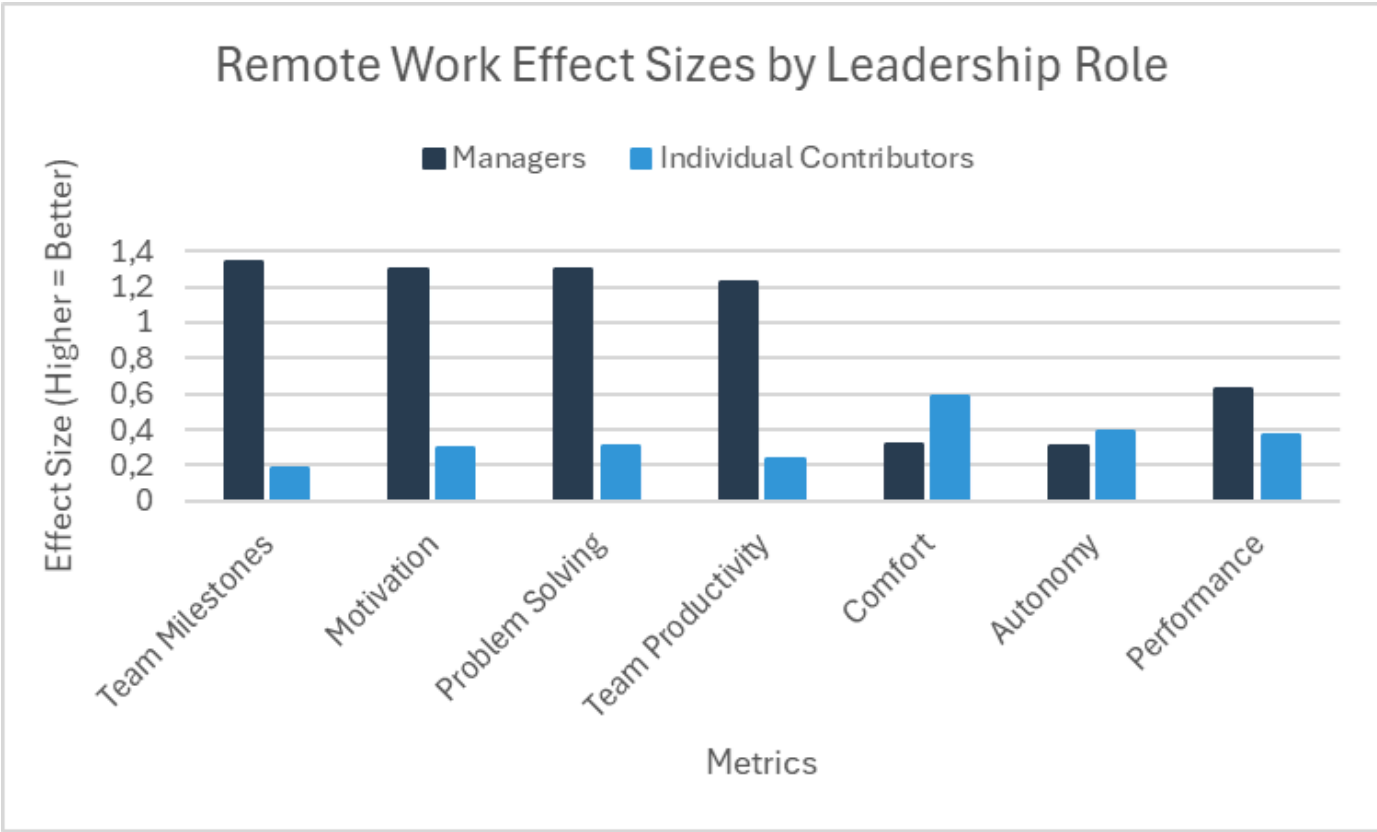


Table 1. Remote Work Effects by Gender-Role Groups

Group	Performance	Team Productivity	Motivation
Male Managers	+0.97*	+1.47*	+1.25*
Male Individual Contributors	+0.48*	+0.40*	+0.35
Female Managers	Insufficient Data	Insufficient Data	Insufficient Data
Female Individual Contributors	+0.13	-0.22	+0.34

*Statistically significant at $p<0.05$ | Color intensity indicates effect magnitude | Green = Positive effects, Red = Negative effects, Gray = Insufficient data

Practical Recommendations

- 3-4 remote days/week for managers, 2-3 remote days/week for individual contributors, with enhanced coordination support for non-managerial roles.
- Develop targeted support structures including networking opportunities for female developers and mentoring partnerships with successful managers to enhance collaborative effectiveness and address gender productivity disparities.